

Research Proposal: Secure and Trustworthy Data Sharing for Maritime AI using Federated Learning and Physics-Informed Digital Twins

1. Introduction and Research Questions

The digitalization of the maritime industry hinges on the deployment of robust Artificial Intelligence (AI) models. However, maritime stakeholders operate in highly competitive and geopolitically sensitive environments. Consequently, the operational data required to train these AI models (e.g., sensor logs, ship design parameters, routing data) is fiercely protected. The central challenge of this PhD project is how to train global, highly accurate maritime AI models without compromising the data sovereignty, cybersecurity, and confidentiality of the contributing partners.

This research proposal addresses the following core research questions: - **RQ1:** How can we construct a decentralized data-sharing architecture that allows maritime stakeholders to collaboratively train AI models without exchanging raw, sensitive operational telemetry? - **RQ2:** How can physics-informed constraints (e.g., fluid dynamics, thermodynamics) be integrated into decentralized AI models to detect adversarial data poisoning and cyber-anomalies? - **RQ3:** What is the quantifiable trade-off between cryptographic security overhead (e.g., homomorphic encryption, differential privacy) and the real-time inference latency required for maritime operations?

2. Theoretical Perspectives

This research draws upon three primary theoretical domains:

2.1. Information Security & Federated Learning (FL)

Rather than aggregating data in a centralized data lake—which creates a massive cybersecurity single point of failure—this project proposes utilizing Federated Learning. In FL, the AI model is distributed to the stakeholders’ edge devices (e.g., on-board ship servers). The model trains locally on the sensitive data, and only the mathematically encrypted weight updates (gradients) are transmitted back to the Norwegian Maritime AI Center. This guarantees data sovereignty, as raw data never leaves the ship.

2.2. Physics-Informed Neural Networks (PINNs) as Anomaly Detectors

A critical vulnerability in decentralized AI is “data poisoning,” where a malicious or compromised node injects fake gradients to corrupt the global model. Drawing from my previous engineering work on Gas Turbine Digital Twins and Medical Digital Twins, I propose using Physics-Informed Neural Networks. Maritime

vessels are governed by strict physical laws (e.g., Navier-Stokes equations for hydrodynamics). By embedding these differential equations into the AI’s loss function, the central server can automatically reject any gradient updates that violate physical reality, effectively nullifying sophisticated cyber-attacks and sensor spoofing.

2.3. Zero-Trust Architecture

The project will be framed within a Zero-Trust security paradigm, assuming that maritime communication channels (e.g., SATCOM) are inherently vulnerable to eavesdropping and manipulation.

3. Methodological Design

The project will be executed through a mixed-methods approach, combining mathematical algorithm design with empirical testing.

Phase 1: Architecture Design and Simulation

- Formulate the mathematical foundation for a secure Federated Learning protocol tailored to maritime bandwidth constraints.
- Implement secure aggregation techniques (e.g., Multi-Party Computation) to protect the gradients from inference attacks.
- Establish the baseline physics-informed loss functions for a maritime test-case (e.g., condition monitoring of propulsion systems).

Phase 2: Cyber-Anomaly Injection and Detection

- Utilize the CPS Lab at NTNU Ålesund to simulate a distributed fleet of vessels.
- Inject deliberate cyber-attacks into the simulation, including False Data Injection (FDI) and adversarial gradient poisoning.
- Evaluate the efficacy of the physics-informed anomaly detection mechanisms in isolating and ignoring compromised nodes.

Phase 3: Industrial Validation

- Deploy the federated architecture in a closed-loop test with industry partners from the Norwegian Maritime AI Center.
- Measure key performance indicators (KPIs) including model convergence speed, communication overhead, and resilience to simulated network partitions.

4. Progress Plan

- **Year 1:** Conduct systematic literature review on maritime cybersecurity and FL. Formalize the mathematical architecture for physics-informed

federated aggregation.

- **Year 2:** Develop the simulation environment. Conduct stress tests against simulated cyber-attacks. Submit findings to a top-tier cybersecurity/AI conference (e.g., IEEE S&P, NeurIPS).
- **Year 3:** Collaborate with maritime user partners (e.g., SINTEF) to deploy a prototype on real-world edge devices. Focus on optimizing the security-latency trade-off. Publish in high-impact journals (e.g., IEEE Transactions on Information Forensics and Security).
- **Year 4:** Finalize the doctoral dissertation, disseminate findings to the Maritime Forum and Blue Maritime clusters, and complete all remaining coursework.